



U.S. Department
of Transportation

**Pipeline and Hazardous
Materials Safety
Administration**

January 24, 2022

1200 New Jersey Avenue, SE
Washington, DC 20590

DOT-SP 13583
(ELEVENTH REVISION)

EXPIRATION DATE: 2025-12-31

(FOR RENEWAL, SEE 49 CFR 107.109)

1. GRANTEE: Structural Composites Industries (SCI)
Pomona, CA
2. PURPOSE AND LIMITATION:
 - a. This special permit authorizes the manufacture, marking, sale and use of non-DOT specification composite cylinders conforming to all requirements of the DOT-CFFC (basic requirements for fully wrapped carbon-fiber reinforced aluminum liner) cylinder as specified herein, for the transportation in commerce of the materials authorized by this special permit. This special permit provides no relief from the Hazardous Materials Regulations (HMR) other than as specifically stated herein. The most recent revision supersedes all previous revisions.
 - b. The safety analyses performed in development of this special permit only considered the hazards and risks associated with transportation in commerce.
 - c. In accordance with 49 CFR 107.107(a) party status may not be granted to a manufacturing permit. These packaging may be used in accordance with 49 CFR 173.22a.
3. REGULATORY SYSTEM AFFECTED: 49 CFR Parts 106, 107 and 171-180.
4. REGULATIONS FROM WHICH EXEMPTED: 49 CFR §§ 173.302a (a) (1), 173.304a (a) (1), 175.3 and 180.205, in that a non-DOT specification cylinder is not authorized, except as specified herein.
5. BASIS: This special permit is based on the application of Structural Composites Industries (SCI) dated November 19, 2021, submitted in accordance with § 107.109.

6. HAZARDOUS MATERIALS (49 CFR 172.101):

Hazardous Materials Description			
Proper Shipping Name	Hazard Class/ Division	Identification Number	Packing Group
Air, compressed (containing up to 39% by volume oxygen)	2.2	UN1002	N/A
Argon, compressed	2.2	UN1006	N/A
Carbon dioxide	2.2	UN1013	N/A
Helium, compressed	2.2	UN1046	N/A
Hydrogen, compressed	2.1	UN1049	N/A
Methane, compressed or Natural gas, compressed (<i>with high methane content</i>)	2.1	UN1971	N/A
Neon, compressed	2.2	UN1065	N/A
Nitrogen, compressed	2.2	UN1066	N/A
Nitrous oxide	2.2	UN1070	N/A

7. SAFETY CONTROL MEASURES:

a. PACKAGING: Packaging prescribed is a non-DOT specification composite cylinder as described in SCI application on file with the Office of Hazardous Materials Safety (OHMS).

The cylinder must meet all the requirements of the ISO Standard 11119-2: 2002 (Gas Cylinders of Composite Construction- Specification and Test Methods – Part 2) for unlimited service life. Additionally, the cylinder must meet the following:

- (1) The ratio of burst pressure (P_b) over service pressure (P_s) must be greater than or equal to 3.4 ($P_b/P_s = 3.4$).
- (2) All of the batch inspection and testing must be performed for unlimited service life (e.g., hydraulic pressure test will be 12,000 cycles with test pressure).

- (3) The liner is aluminum alloy 6061-T6 and made in accordance with the DOT-CFFC-6 specification.
- (4) Service pressure may not exceed 345 bar (5000 psi).
- (5) Water volume may not exceed 95 liters (210 lbs).
- (6) Each cylinder must be fitted with a pressure relief device in accordance with § 173.301(f).
- (7) Cylinder valve protection must be in accordance with § 173.301(h).
- (8) Cylinder service pressure is 2/3 of the design test pressure.

b. TESTING: The cylinders must be tested in accordance with the ISO 11119-2 prototype design requirements except that:

Section 8.2.1 Cylinder lot for qualification testing shall represent the minimum number of cylinders needed to complete testing. Spare cylinders may be included in the lot at the manufacturer's discretion.

Section 8.2.2 Not applicable.

c. MARKING:

- (1) Each cylinder must be permanently marked (other than by stamping) in the composite on the sidewall. The marking must be easily visible and must be protected from external damage due to the environment and handling.
- (2) The marking must contain the following:
 - (i) DOT special permit number (DOT SP-13583) followed by service pressure expressed in bar (psi).
 - (ii) A serial number and the manufacturer's identification number or a symbol as obtained from the Associate Administrator for Hazardous Materials Safety, located just below or immediately following the DOT marking above. The serial number and the manufacturer's identification number may be placed on the boss provided the marking is accessible for inspection.
 - (iii) The DOT inspector's official mark must be placed near the serial number. The marking must contain date the (month and year) of the initial hydraulic proof pressure test for that cylinder.

(iv) The size of the letters and numbers used must be at least 0.64 cm (1/4 inch) high if space permits.

(v) The following are examples of an authorized format for marking:

DOT-SP 13583-345 bar (5000 psi)
1234-MMI (or symbol)
II—MM/YY

(vi) Additional markings are permitted in the composite, provided the additional markings do not obscure the required marking and are not detrimental to the integrity of the cylinder.

(vii) Provisions for marking of the required requalification dates and RIN information must be made near the cylinder markings.

d. Requalification: Each cylinder must be requalified once every 5 years by a qualified person holding a valid DOT RIN using a hydraulic proof pressure test equal to 1.5 times the marked service pressure and hold the pressure for a minimum of 3 minutes without a loss of pressure. Additionally, each cylinder must visually be inspected in accordance with CGA Pamphlet C-6.2 Guidelines for Visual Inspection and Re-qualification of Fiber Reinforced High Pressure Cylinders, except as specifically noted herein:

(1) Cylinders with fiber damage (cuts, abrasions, etc.) that exceeds Level 1 type damage as defined in CGA Pamphlet C-6.2 and meet the following depth and length criteria are considered to have Level 2 damage:

(a) Depth - Damage that upon visual inspection is seen to penetrate the outer fiberglass layer but does not expose the carbon layer beneath, or that has a measured depth of greater than 0.005 inch and less than 0.045 inch for cylinders with an outside diameter greater than 7.5 inches or less than 0.035 inch for cylinders 7.5 inches or less in outside diameter;

(b) Length - Damage that has a maximum allowable length of:

Region	Direction of fiber damage	Maximum length of damage
Cylinder sidewall and domes	Transverse to fiber direction (longitudinal direction)	20% of the straight sidewall section length
Cylinder sidewall and domes	In fiber direction (circumferential direction)	20% of the straight sidewall section length

(2) Cylinders with damage that meet the Level 2 criteria must be rejected. Requalifiers must contact the cylinder manufacturer in the event that the damage cannot be clearly interpreted based on these criteria. Repair of rejected cylinders is authorized for Level 2 type damage. Repairs must be made in accordance with CGA Pamphlet C-6.2, prior to the hydrostatic pressure test. Repairs must be evaluated after the hydrostatic test.

(3) Cylinders that have direct fiber damage that penetrates through the outer fiberglass layer and into the carbon layer, or that have a measured damage depth of greater than the Level 2 maximum are considered to have Level 3 type damage.

Cylinders that have damage with depth meeting Level 2 but length exceeding the Level 2 maximum is considered to have Level 3 type damage. Cylinders with Level 3 type damage are not authorized to be repaired, and must be condemned.

(4) Cylinders that are designed and manufactured with protective boot must be rejected or returned to manufacturers for repair under the following condition:

- (i) Boot is missing;
- (ii) Boot has damaged that allows the composite surface to be visible (exposed) through damage (e.g., crack, gaps, and holes).

(5) A hydrostatic requalification may be repeated as provided in § 180.205(g); only two such tests are permitted. Pressurization prior to the official hydrostatic test for the purpose of a systems check may not exceed 85% of the minimum required test pressure.

(6) Persons who perform inspection and testing of cylinders subject to this special permit must comply with § 180.205(b) and with all the terms and conditions of this special permit.

(7) Requalification date (month/year) must be permanently marked on the cylinder as specified in paragraph § 180.213. The marking of the RIN symbol on the cylinder certifies compliance with all of the terms and conditions of this special permit.

e. OPERATIONAL CONTROLS:

(1) Cylinders manufactured under this special permit are authorized for a maximum service use of 30 years from the date of manufacture, provided the manufacturer meets all requirements specified under paragraph 8.a. of this special permit.

(2) Cylinders may not be used for underwater breathing purposes.

(3) Cylinders used in oxygen service must conform with § 173.302(b)(1) through (4).

(4) Cylinders used in nitrous oxide service must conform with § 173.304a(a) and meet the valve and cleaning requirements in § 173.302(b).

(5) A cylinder that has been subjected to fire may not be returned to service.

(6) Transportation of Division 2.1 (flammable gases) are not authorized aboard cargo vessel or aircraft unless specifically authorized in the Hazardous Materials Table (§ 172.101).

(7) Cylinders must be packaged in accordance with § 173.301(a)(9).

(8) Design qualification and batch (lot) testing must be performed on a completed cylinder. Addition of a boot to the cylinder is not authorized during the drop test unless it is a permanent part of the cylinder.

8. SPECIAL PROVISIONS:

a. Recall Program:

(1) SCI must formulate a recall program for service life extension within 12 months of the issuance of this special permit. The recall program must include detail procedure for obtaining the cylinders from field and prototype design qualification testing of each design type.

(2) SCI must randomly recall a minimum of thirty cylinders of each design type which have been in service for 10 and 15 years. Cylinders recalled after 10 years shall be designated "Group A" and cylinders recalled after 15 years shall be

designated “Group B”. All recalled cylinders must be requalified as specified in paragraph 7. (c) above. All cylinders that fail to meet the requalification requirements must be condemned, removed from service and rendered incapable of retaining pressure. In the case that some units from the initial minimum lot size are condemned, additional cylinders must be selected and requalified as specified in paragraph 7.(c), above, until a group of 30 cylinders has been collected. These 30 cylinders constitute a batch for additional testing. The cylinders must be tested in accordance with Sections 8.5.4, 8.5.5, 8.5.7 and 8.5.8 of ISO 11119-2:2002. An Independent Inspector must witness all tests. Acceptance criteria shall be as defined in ISO 11119-2:2002 except $P_b = 1.8P_h$ and y must be greater than or equal to 20.

(3) The complete test report including original test data must be submitted to the Associate Administrator for Hazardous Materials Safety for assessment within 30 days of completion. Failure to meet the acceptance criteria specified in this section shall result in the design being restricted to a maximum life of 15 years.

- b. In accordance with the provisions of Paragraph (b) of § 173.22a, persons may use the packaging authorized by this special permit for the transportation of the hazardous materials specified in paragraph 6, only in conformance with the terms of this special permit.
- c. A person who is not a holder of this special permit, but receives a package covered by this special permit, may reoffer it for transportation provided no modification or change is made to the package and it is offered for transportation in conformance with this special permit and the HMR.
- d. A current copy of this special permit must be maintained at each facility where the package is offered or reoffered for transportation.
- e. A current copy of this special permit must be maintained at each facility where the package is manufactured under this special permit and must be made available to a DOT representative upon request.
- f. Each packaging manufactured under the authority of this special permit must be either: (1) marked with the name of the manufacturer and location (city and state) of the facility at which it is manufactured; or (2) marked with a registration symbol designated for a specific manufacturing facility by the Office of Hazardous Materials Special permits and Approvals for a specific manufacturing facility.
- g. The cylinders described in this special permit are authorized only for normal transportation as an article of commerce i.e., the movement of hazardous materials packages from consignor to consignee.

- h. When authorized for transportation by cargo vessel as prescribed in § 172.101, Hazardous Materials Table, flammable gases covered by this special permit must be packed within a closed freight container of steel construction.
 - i. Transportation of oxygen is only authorized when in accordance with § 175.501.
9. MODES OF TRANSPORTATION AUTHORIZED: Motor vehicle, rail freight, cargo vessel, passenger-carrying aircraft, and cargo only aircraft (see paragraph 8.i. for restrictions).
10. MODAL REQUIREMENTS: A current copy of this special permit must be carried aboard each cargo vessel or aircraft used to transport packages covered by this special permit. The shipper must furnish a current copy of this special permit to the air carrier before or at the time the shipment is tendered.
11. COMPLIANCE: Failure by a person to comply with any of the following may result in suspension or revocation of this special permit and penalties prescribed by the Federal hazardous materials transportation law, 49 U.S.C. 5101 et seq:
- o All terms and conditions prescribed in this special permit and the Hazardous Materials Regulations, Parts 171-180.
 - o Persons operating under the terms of this special permit must comply with the security plan requirement in Subpart I of Part 172 of the HMR, when applicable.
 - o Registration required by § 107.601 et seq., when applicable.

Each AHazmat employee@, as defined in § 171.8, who performs a function subject to this special permit must receive training on the requirements and conditions of this special permit in addition to the training required by §§ 172.700 through 172.704.

No person may use or apply this special permit, including display of its number, when this special permit has expired or is otherwise no longer in effect.

Under Title VII of the Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (SAFETEA-LU) - “The Hazardous Materials Safety and Security Reauthorization Act of 2005” (Pub. L. 109-59), 119 Stat. 1144 (August 10, 2005), amended the Federal hazardous materials transportation law by changing the term “exemption” to “special permit” and authorizes a special permit to be granted up to two years for new special permits and up to four years for renewals.

12. **REPORTING REQUIREMENTS:** Shipments or operations conducted under this special permit are subject to the Hazardous Materials Incident Reporting requirements specified in 49 CFR §§ 171.15 - Immediate notice of certain hazardous materials incidents, and 171.16 - Detailed hazardous materials incident reports. In addition, the grantee(s) of this special permit must notify the Associate Administrator for Hazardous Materials Safety, in writing, of any incident involving a package, shipment or operation conducted under terms of this special permit.

Issued in Washington, D.C.:



for William Schoonover
Associate Administrator for Hazardous Materials Safety

Address all inquiries to: Associate Administrator for Hazardous Materials Safety, Pipeline and Hazardous Materials Safety Administration, U.S. Department of Transportation, East Building PHH-13, 1200 New Jersey Avenue, Southeast, Washington, D.C. 20590.

Copies of this special permit may be obtained by accessing the Hazardous Materials Safety Homepage at <https://www.phmsa.dot.gov/approvals-and-permits/hazmat/special-permits-search>. Photo reproductions and legible reductions of this special permit are permitted. Any alteration of this special permit is prohibited.

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